… Lecture 9 …
Variances and Perspectives

Part 1:
Comparing genetic variances: what is Vg?

There are two parts to the lecture: firstly, looking at the interpretation of the estimated variance components and heritabilities; secondly, a look at what factors determine the accuracy of genomic models.

Estimates of Heritability

- We know what we mean by $h^2 = V_A / V_P$
  - » e.g. $h^2 = 0.5$ for milk yield

Are we sure?

- Heritability varies across populations
  - » e.g. breeds
  - » … and across herds
  - » … and within families
- We have an idea it refers to 'some' population
  - » which?

The term 'heritability' is often used and often estimated, and estimates are often compared – but are we always clear on what it is meant? For example, it is stated $h^2 = 0.5$ (say) for milk yield, or 0.8 for human height, and it is understood that a calculation of V_A/V_P is being made. However, given some consideration, we might expect h^2 to vary across breeds, or across sub-populations such as ethnic groups. Therefore, if an estimate is made of V_A, what population does it refer to?

Ancient History

- Pre-REML
  - » heritability estimated from selection experiments
  - » variance of paternal half-sibs (PHS)
    - » Sire Models!
    - » important for good estimates in real populations
  - » sires often selected before breeding
    - » estimates are biased!
    - » what are we estimating?

For milk yield, prior to BLUP and REML, h^2 was often estimated from selection experiments as a response to selection intensity applied, or from variances of a sample of PHS families using sire models, maybe from a selected sample of sires. This would lead to biased estimates.
With the implementation of BLUP and REML, and the routine use of the **A** matrix, there was some greater clarity on what h^2 might refer to.

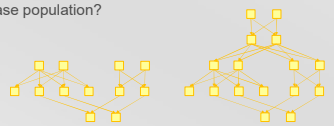
REML Age

- Estimate of V_A refers to the base population
  - » i.e. the sub matrix of **A** which is identity matrix
    - » but are they truly unrelated individuals!
  - » what base population?

The estimate of additive genetic variance V_A from REML using **A** is an estimate of the variance of breeding values <u>in the base population</u>. While the estimate of residual variation V_E is readily interpretable as what is not explained by our genetic model (or other terms), this estimate of V_A from REML is not straightforward – are the base population unrelated (more realistically, to what degree are they related).
One consequence of the implementation of REML was the estimate of h^2 for milk yield increased steadily (from $h^2 \approx 0.25$) as the estimate shifted from being derived from a selected sample of sires, to a base population with selection accounted for, as far as possible.
Furthermore, if the recorded data remains the same, and some additional ancestral pedigree is inserted then the estimate of V_A will change – so the same data will give a range of estimates of V_A and so of h^2.
Here is an extreme example of adding some extra pedigree – a couple of generations of full-sib mating.

REML Age

- Estimate of V_A refers to the base population
  - » i.e. the sub matrix of **A** which is identity matrix
    - » but are they truly unrelated individuals!
  - » what base population?



Genomic Age

- Van Raden **G** … estimates V_G
  - » genic variances - all loci in LE and in HWE
  - » individuals used to construct **G** if markers in HWE
  - » can you compare with V_A
- Change **G** … there are many!
  - » e.g. prune a few SNPs
  - » does estimate of V_G refer to same population?
  - » is h^2 same?

This is now more problematic with the genomic age. The genetic variance estimated by Van Raden matrices is a genic variance refer either to a population with genomic relationship given by **I**, which would be expected if the loci are in linkage equilibrium and in Hardy-Weinberg equilibrium (HWE). It can be shown that if the marker loci are (on average) in HWE for the individuals represented in the **G** matrix then the variance parameter is an estimate of the additive genetic variance in that set of individuals. However, can this parameter be compared to the additive genetic variance from **A**? Yet many paper do make this comparison without question. Further what happens if there is no HWE?

Perspectives

- Specific populations and sub-populations of interest …
  - » e.g.
    - » all individuals with data
    - » particular generations
    - » selection candidates
- Can compare different models:
  - » accuracies
  - » also predicted variances

Beyond this – how might we estimate the genetic variance in a particular sub-population of interest: e.g. individuals with data, or a particular generation which may be selection candidates, or a subgroup defined by geography or culture?
Maybe the genetic variance is estimated by different models – do they give notably different estimates?

General Result

- For a population with (co)variance matrix **V**
  - » can refer to a trait phenotype, breeding value, …
  - » individuals have value y_i
  - » $v_{ii} = \text{var}(y_i)$, $v_{ij} = \text{cov}(y_i, y_j)$
- Variance within the population

$$= E[(y_i - \bar{y})^2] = E[y_i^2] - E[\bar{y}]^2$$

  - » a standard result for variances

$$= \overline{diag(\mathbf{V})} - \bar{\mathbf{V}}$$

- See handout

So start with a general result. Let a set of n individuals have a (co)variance matrix **V**, it could be $V_a\mathbf{A}$ or $V_g\mathbf{G}$, then the variance *within* the set (i.e. $E[(g-\bar{g})^2]$) is given by $\overline{diag(\mathbf{V})} - \bar{\mathbf{V}}$ (i.e. the mean of the leading diagonal elements minus the mean of all the elements of **V**. This is derived in more detail in the handout.

As an example, $\begin{pmatrix} 3 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 4 \end{pmatrix}$, the variance within the subset is simply derived as 9/3 – 9/9 = 2.

Apply to Relationships

- **A** matrix
  - » (co)variance of breeding values is $V_A\mathbf{A}$
  - » sub-matrix for base generation of size n is $V_A\mathbf{I}_n$
  - » so variance within the base is $V_A(1-n^{-1})$
    - » as $\overline{\text{diag}(\mathbf{I}_n)} = 1$ and $\bar{\mathbf{I}}_n = n/n^2 = n^{-1}$
- So the REML estimate of parameter $\hat{V}_A$ is estimating variance for a *very large* base population from which n sampled
  - » e.g. variance within this base is $\frac{3}{4}\hat{V}_A$

Consider an **A**-matrix with n individuals in the base with the variance component estimated as $\hat{V}_a$ then the co(variance) matrix of the base is $\hat{V}_a\mathbf{I}_n$ and so, applying the formula, the variance *within* the base is $(1-n^{-1})\hat{V}_a$. So REML is estimating the variance for a very large base population from which n are sampled.
In the pedigree shown in the slide the variance within the population is $\frac{3}{4}\hat{V}_a$ as there are 4 individuals in the base.

Apply to Relationships

- Van Raden $\mathbf{G}_1$ or $\mathbf{G}_2$
  - » (co)variance of breeding values is $V_G\mathbf{G}$
  - » $\bar{\mathbf{G}} = 0$ as sum of elements is 0
  - » if all loci are in HWE …
    - » $diag(\mathbf{G}) = 1$
    - » so REML estimate $\hat{V}_G$ is estimating genetic variance within set used to construct **G**
  - » see handout 'properties of Van Raden matrices'

For a Van Raden $\mathbf{G}_1$ or $\mathbf{G}_2$, then $\bar{\mathbf{G}} = 0$. If all the loci are then in HWE, or more generally if $\overline{diag(\mathbf{G})} = 1$ then $\hat{V}_g$ is the estimate of genetic variance *within* the set of individuals used to construct **G**. Note it is individuals used to *construct* **G** irrespective of whether they have phenotype records – the records will influence the parameter estimate but not its interpretation. For example, in cross-validation the **G** does not change with the validation set and so the interpretation as an estimate of variance within the whole set does not change either,

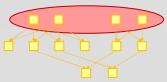
Apply to Relationships

- $\mathbf{G}_1$ matrix … Van Raden 1 for the base

$$V_G(\overline{diag(\mathbf{G}_1[base])} - \overline{\mathbf{G}_1[base]})$$
$$= (0.6625 - 0.0669)V_G$$
$$= 0.60V_G$$

0.27	-0.42	-0.42	0.22
-0.42	0.79	0.79	-0.48
-0.42	0.79	0.79	-0.48
0.22	-0.48	-0.48	0.80

c.f. $0.75V_A$!



Applying the formula to our base generation from the toy pedigree in earlier lectures: with $\mathbf{G}_1$ the estimate of genetic variation within the base is $0.60V_g$ – compared to $0.75V_a$ when derived from the **A**-matrix.

The scaling factor k can be very different from 1 if (i) the number of individuals is small, or (ii) if the population is not in HWE when using Van Raden matrices.

Conclusion

1. Estimates of V_A & V_G from different relationship matrices not directly comparable!
2. Compare models by:
   - a) defining a subset of interest
     - i. e.g. whole population
     - ii. e.g. genotyped individuals
   - b) estimate variance, estimate h^2 etc. on same subset for each relationship matrix

Estimates of genetic variance derived from using **A** and **G** are not directly comparable. However, one can compare models by defining a subset of interest and looking at the estimates of variance within the subset.

Comparing Genetic Variances

- Define subset S … e.g. for **A**
  - » find submatrix **A**[S]
  - » take average of diagonal of **A**[S]
  - » take average of all **A**[S]
  - » find difference, say *k*
  - » estimated variance within S is $k\hat{V}_A$
  - » s.e. is $k\ s.e.(\hat{V}_A)$
  - » can calculate h^2[S]
    - » e.g. vpredict(my.asr, hh[S] ~ k*V1/(k*V1+V2))

Strictly V2 should also be replaced by $(1-n^{-1})V_2$ where n is size of S

The general formula can be applied to any subset S of the individuals appearing in *any* relationship matrix, e.g. **A** with a variance parameter estimate $\hat{V}_a$:

(i) identify the sub-matrix for S, **A**[S];

(ii) take average of the elements on the leading diagonal of the sub-matrix, and all the elements, and let *k* be the difference;

(iii) the estimate of variance *within* S is $k\hat{V}_a$;

(iv) the s.e. of the estimate is $k\ s.e.(\hat{V}_a)$ – if I scale any parameter's estimate by *k*, the s.e. is scaled by the same amount!

(v) In a simple 2 component model, the heritability *within* the subset S is $h^2 = k\hat{V}_a / (k\hat{V}_a + \hat{V}_e)$.

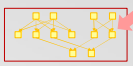
Strictly the same idea of calculating the variance *within* S should be applied to all the variances, so $\hat{V}_e$ should be scaled by 1-1/*n* where *n* is the number in the set S. When *n* is large, e.g. >1000, this correction is obviously small.

Example

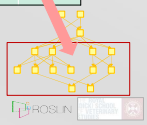
In table below V_A is the component, **V** is $\mathbf{G}_1$, **A**, $\mathbf{G}_{FG}$, or $\mathbf{A}_+$

Matrix	Analysis V_A	Analysis s.e.(V_A)	$\overline{diag(\mathbf{V})} - \overline{\mathbf{V}}$	Target Population V_A	Target Population s.e.(V_A)
$\mathbf{G}_1$					
A					
$\mathbf{G}_{FG}$					
$\mathbf{A}_+$					

$\mathbf{G}_1$, **A**, $\mathbf{G}_{FG}$
Pedigree has data and genotypes



$\mathbf{A}_+$
Data and genotypes as for $\mathbf{G}_1$, **A**, $\mathbf{G}_{FG}$ plus additional pedigree



Look at the set of 12 individuals used in Lecture 8 assumed to have data and genotypes. Consider also a slight modification by adding 4 ancestors to the pedigree that have ***no*** phenotype or genotype data.

The genetic variance was estimated in linear mixed models using **A**, $\mathbf{G}_1$ and $\mathbf{G}_2$ based on the 12 individuals with (simulated) records, and $\mathbf{A}_+$ the extended pedigree of 16 individuals with only the same 12 individuals with the same records. Remember the estimated component for **A** and for $\mathbf{A}_+$ refer to their respective bases.

Example

In table below V_A is the component, **V** is $\mathbf{G}_1$, **A**, $\mathbf{G}_{FG}$, or $\mathbf{A}_+$

Matrix	Analysis V_A	Analysis s.e.(V_A)	$\overline{diag(\mathbf{V})} - \overline{\mathbf{V}}$	Target Population V_A	Target Population s.e.(V_A)
$\mathbf{G}_1$	14.6	10.5	0.973	14.2	10.2
A	22.2	19.5	0.698	16.3	13.6
$\mathbf{G}_{FG}$	24.4	12.5	0.701	17.1	8.8
$\mathbf{A}_+$	38.0	32.1	0.422	16.0	13.5

$\mathbf{G}_1$, **A**, $\mathbf{G}_{FG}$
Pedigree has data and genotypes

$\mathbf{A}_+$
Data and genotypes as for $\mathbf{G}_1$, **A**, $\mathbf{G}_{FG}$ plus additional pedigree

The table shows the component that was estimated using each of the models. One can see that there is nearly a 3-fold difference in the estimates (albeit with large s.e.s!). It is also clear that $\overline{diag(\mathbf{V})} - \overline{\mathbf{V}}$ differs widely with the $\mathbf{A}_+$ estimate having both the largest variance estimate and the smallest k. When scaled to estimate the population within the set of 12 individuals then the variances are not identical (they have different models for the breeding values) but are much more similar.

The example above is small, so s.e.s are large, but one hopes it is illustrative.

Remember

- The best model is one that predicts well
  - » not the one with the highest h^2

As a final point remember the best model is the one that gives the best predictions, not the one with the highest heritability.

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Variances and Perspectives

Part 2:
A is from Ridge Regression too!

A theme present in the lectures is a different approach to estimating breeding values and additive merits when using the genomic data. It is common to encounter the idea that the properties of an **A**-matrix should be evident somewhere in in $\mathbf{G}_1$ or $\mathbf{G}_2$ based on an assumption that a numerator relationship matrix is simply the expectation of the genomic **G**, and therefore strengthening any correspondence will improve the performance of the relationship matrix. Beware, the origins of the matrices come from different underlying models and this section makes the differences clearer.

Van Raden G & $\mathbf{G}_2$

Classical ridge regression assumptions

- Deri
- $\mathbf{G}_1$ derived via SNP BLUP 1: $\mathbf{y} = \ldots + \mathbf{W}_1\mathbf{b} + \ldots$
  - » **b** is (*m* x 1) vector of random marker effects
  - » **b** assumed to be MVN(0,σ_b^2**I**)
  - » covariates $\mathbf{W}_1$ are allele counts 0,1,2
- $\mathbf{G}_2$ derived via SNP BLUP 2: $\mathbf{y} = \ldots + \mathbf{W}_2\mathbf{b} + \ldots$
  - » **b** is (*m* x 1) vector of random marker effects
  - » **b** assumed to be MVN(0,σ_b^2**I**)
  - » covariates $\mathbf{W}_2$: locus *j* counts scaled by $\sqrt{2p_j(1-p_j)}$

The Van Raden matrices have been introduced in this course as ridge regressions. The key assumptions is an underlying model of the breeding values that can be formulated as a multiple regression with effects assumed to be distributed MVN(0,**I**). The relationship matrices then emerge by considering the (co)variances of the breeding values implied by the model.

A as a Ridge Regression

- **A** derived from regression: $\mathbf{y} = \ldots + \mathbf{Rb} + \ldots$
  - » where **b** is *p* x 1 vector of effects assumed MVN(0,σ_b^2I)
  - » *p* is the number in pedigree
  - » with effects $A_i/\sqrt{2}$ for i in base and $a_i\sqrt{(1-F_i)}$ if not
  - » σ_b^2 is ½$V_{A,0}$
  - » covariates **R** are scaled contributions to descendants
    - » analogous to the allele counts in SNP BLUP

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The **A** matrix was originally introduced from probabilities of IBD and coefficients of kinship. It can also be derived as a ridge regression by:
(i) covariates as contributions of the base generation individuals to descendants scaled by √2, and contributions of all other individuals to descendants scaled by 1/√(1-F), so the number of covariates is equal to the number of individuals in the pedigree;
(ii) effects as $A_i/\sqrt{2}$ for the base generation and $a_i\sqrt{(1-F)}$ for all other individuals, where A_i is the breeding value and a_i is the Mendelian sampling term;
(iii) the effects will then be distributed as MVN(0,**I**).
Therefore, the model based on IBD leading to **A** is a regression on contributions with effects being breeding values of base individuals and Mendelian sampling terms of all the other individuals. The scaling of covariates and effects in (i) and (ii) above is simply to make the coefficients have the form of the classical ridge regression with distribution MVN(0,**I**)!

With IBD

Strictly additive traits

Infinitesimal Model

... ridge regression on scaled base breeding values & Mendelian sampling terms

Measures of change in diversity from to a reference base

F_{IBD} F_{drift} F_{het} F_{var}

Covariance of breeding values i.e. numerator relationships a_{ij}

IBD Relationship Matrix
$\mathbf{g} \sim MVN(0, V_a\mathbf{A})$

F_{IBD} F_{drift} F_{var} F_{het}

A A = 2K **K** Kinship coefficients ϕ_{ij}

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Using the model of relationship based on IBD, the measures of inbreeding based on IBD, drift or heterozygosity coincide.

With IBS

Strictly additive traits

Model of gene/marker effects e.g.

... ridge regression on allele counts V_aI)

Measures of change in diversity from to a reference base

F_{IBD} F_{drift} F_{het} F_{var}

Covariance of breeding values i.e. relationships g_{ij}

Relationship matrices e.g. Van Raden matrices

e.g. $\mathbf{G_1}$

F_{IBD} F_{var} F_{drift} F_{het}

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Using the model of relationship based on IBS, measures of inbreeding using concepts of drift or heterozygosity (or IBD) no longer coincide and can diverge. $\mathbf{G_1}$ is constructed using deviations from mean allele frequency and uses the concept of drift from a reference point. The genomic relationship $\mathbf{G}_{FG}$ is constructed to track IBD from a base. Other **G** can be constructed to use the idea of heterozygosity as a reference.
See *Frontiers in Genetics* **11**:880 for further (optional) information.

Summary

- Demonstrates
  - » different perspectives of IBD & IBS models
  - » different perspectives have consequences
  - » both coherent
  - » both reasonable in their own context

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See the optional handout exploring Principal Components in more detail – the idea is used in many applications.

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Part 3: A quick look at reducing dimensionality

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Predictors >> Data

- An increasingly common problem
  - » e.g. 100,000s markers, 1000's phenotypes
  - » 'omics data generates yet more 100,000s of expression data to fit the same 'few' phenotypes!
- Options
  - » shrinkage
  - » reduce dimensionality
  - » subset selection

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When introducing potential methods to cope the vast numbers of SNPs and small numbers phenotypes there were other ideas such as reducing the dimensionality of the SNP data or selecting a subset of SNP and not using those excluded.

Reducing dimensions use the general approach of singular value decomposition. Two methods can be identified Principal Components or Partial Least Squares.

Principal Components

- Marker combinations with greatest variation
  - » i.e. principal components
    - » see handout
    - » linear combinations of allele counts of all loci
  - » orthogonal
  - » ordered to have diminishing variance
  - » 'directions' that are most informative
    - » best estimated
    - » **not necessarily** most important for trait
- Singular Value Decomposition of **W** matrix

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Principal Components finds linear combinations of the allele counts of the genotyped loci (components) that have the greatest variances in the genotyped population.
This can be achieved by the 'Singular Value Decomposition' of the **W** matrix (not part of course!). This decomposition finds the linear combinations of SNP counts step-by-step from the greatest variance, progressing down to the combination with the least, with each combination 'orthogonal' (independent) to all the previous ones.

Principal Components

- Have minimum(m,n) components
- Assume first t explain most variation in phenotypes!
  - » i.e. as well as variation in markers
  - » choose t < minimum (m,n)
  - » by validation
- Treat as random effects and approximate G
  - » reduce computation
- Fit as fixed effects

m = # markers
n = # individuals

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The number of components possible is the minimum of the number of individuals and the number of markers. They can be fitted as fixed or by ridge regression.
The regression coefficient of the phenotypes on the first component (as x-variate, and which has greatest variance) is estimated most precisely as the s.e. of a regression coefficient is $\sim 1/(n\,\mathrm{var}(x))$ where n is the number of points. The regression on the second component will be the next most precise, and so on. Since the component itself is derived independently from the phenotypes, it says nothing about whether the magnitude of the coefficient obtained is big or small, only that it is well estimated! It is often assumed that the initial components will capture the most variation in the phenotypes related to the x-variates (i.e. SNP genotypes) as variance explained by regression is $\sim b^2 n\,\mathrm{var}(x)$ where *b* is the coefficient and *var(x)* is the variance of the *x*-variate in the regression. Therefore the approach relies on reducing the x-data to a smaller number of dimensions by only using the first *t* components, taking identifying directions in which *var(x)* is large, and not necessarily *b* being large.

Partial Least Squares

- Another procedure to reduce dimensionality
- Uses (co)variances of **y** with covariates $\mathbf{W}_2$
- Set of components
  - » c.f. Principal Components
- Fit reduced number as fixed effects
- All traits may not have same components!
- PLS is a valued technique in some problems

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Partial Least Squares (PLS) is similar in that it finds linear combinations of the counts which combine both high variances in the component (hence precision) and high correlations with the phenotypes (hence relevance to the phenotype). As a technique, PLS is widely used in many prediction settings.

Reducing Dimensions

- Empirically fixed PC regression and PLS not as effective as shrinkage e.g. (e.g. **G**)
  - » see Solberg et al. (2009)
- Renewed can simplify computation of EBV treating PC as random and approximating **G**
  - » i.e. shrinkage, but reduced dimensionality
  - » for computational efficiency with huge data
- Require all the SNP genotypes to obtain PC!

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An important point about Principal Components and Partial Least Squares is that although it reduces the dimensions in the model, it does not automatically reduce the number of SNPs required to be genotyped as each component is a linear combination of all the SNPs available!

Subsets of Markers

- Lasso
- Bayes B … Lecture 10

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Some methods associated with subset selection are Lasso, and Bayes B which is touched upon in Lecture 10.

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Part 4: Factors affecting accuracy

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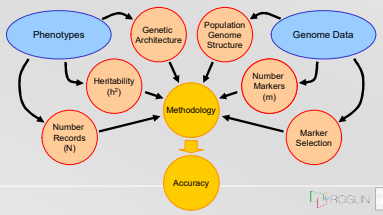
The accuracy is relevant to ranking individuals whether for selection or not, as it is the correlation between the EBV and TBV, and in selection gain is directly related to this accuracy. The bias is the slope of the regression of TBV (on *y*-axis) on EBV (*x*-axis): an unbiased predictor will have a slope 1, if the slope is < 1 then values < mean EBV are expected to increase as more information is obtained, and conversely those EBV > mean EBV are expected to decrease; i.e. the EBV are biased. Likewise, if the slope is > 1, the low EBV are expected to decrease and the high EBV are expected to increase; i.e. the EBV are biased.
Here we shall focus on accuracy and look at what features of the problem may result in obtaining greater accuracy in genomic prediction. For example, how much work might be required to develop a genomic prediction for a new species and trait?

Accuracy Criterion

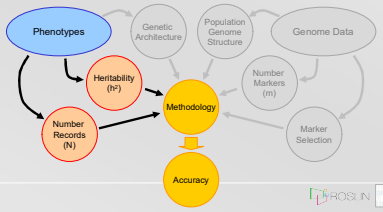
- Accuracy of evaluation, $r_{\hat{g},g}$
- Correlation of $\hat{g}$ with g where
  - » g is true breeding value
  - » $\hat{g}$ is estimate of breeding value from markers
- What is needed to predict accuracy for a new species and trait?

Factors for Accuracy



We can look at this problem as having attributes concerned with phenotypes, genomic data and methodology. These determine the parameters that determine accuracy. In the QGM course, we have concentrated on the 'methodology' that arises from the models.

Factors for Accuracy



Phenotype attributes include number of records, heritability and genetic architecture. For now, just consider the first two of these and the genetic architecture will be looked at in Lecture 10.

Accuracy as f(h^2,N)

- Published formulae predict $r_{\hat{g},g} \sim f(h^2N)$
  - » c.f. shrinkage, amount of genetic information
  - » h^2 richness/record, N number of records

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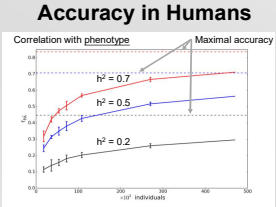
It is natural to think that accuracy would be related the information. The idea of heritability being the richness of genetic information in a record has been introduced previously, so it would follow that accuracy might be a function of h^2N and this has been shown in published formulae.

Accuracy in Humans

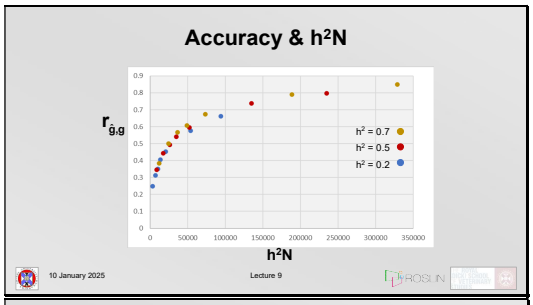


20% QTN included in markers. Simulated from HAPMA2 CEU human population

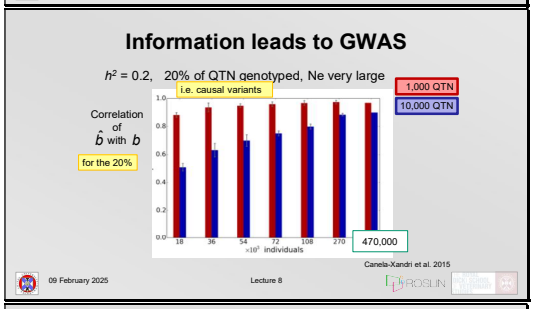
From Canela-Xandri et al. 2015

26 January 2025 Lecture 10

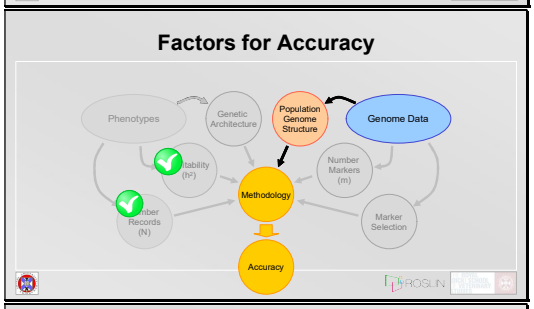
Empirically one can simulate and obtain accuracy as a function of the number of records for a simulation of human genomes, and the graph shows the correlation between $\hat{g}$ and the phenotype found by Canela-Xandri et al. (2015). Note the very large number of records required to approach the maximum accuracy, which is *h*; remember, a perfectly predicted breeding value only predicts the phenotype with a correlation of *h* as the breeding value is assumed independent of the environmental deviation of the phenotype. (The number of SNPs were 590,004.)



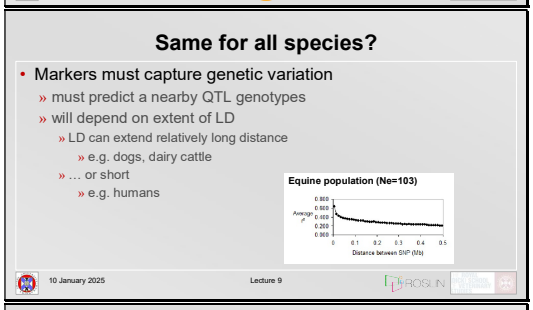
From the simulations, as h^2 is known, one can calculate h^2N, and also scale the observed correlation with phenotype to obtain the accuracy $r_{\hat{g},g}$. The graph is convincing that $r_{\hat{g},g} = f(h^2N)$.



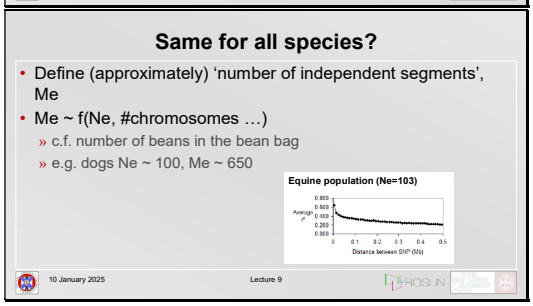
It is worth noting that increasing information 'leads to GWAS'. In the simulations of Canela-Xandri et al. (2015), some of the SNPs used were causal variants and these estimated SNP effects can be correlated with the true effects. It can be seen that as N increases this correlation approaches 1. This is true even when the # variants explaining the variance is very large and therefore the effects are smaller, although the approach to 1 is slower as it it is harder to discriminate such effects from the background.



What about the structure of the genome – which will differ between species, with some species having comparatively long-range LD such as livestock, whereas with humans LD extends over only very short distances. This will typically depend on population history in relation to Ne.



If LD extends for only very short distances, then the genome looks more like many unlinked blocks of loci. In contrast, *and taking it to an extreme*, if substantial LD extends over the whole chromosome, then the number of loci is similar to the number of chromosomes. In the latter case, the variation would then appear to be explained by many fewer parameters of larger effect, which will give a better prediction from the same number of records. This idea can be encapsulated in the idea of 'the number of independent segments' in a genome, often denoted Me.

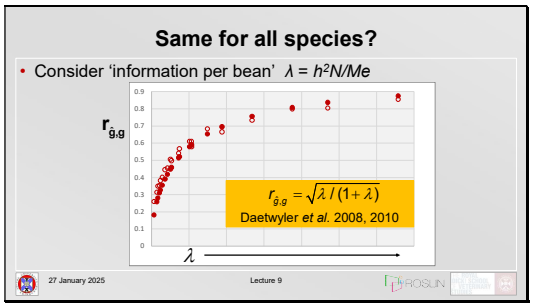


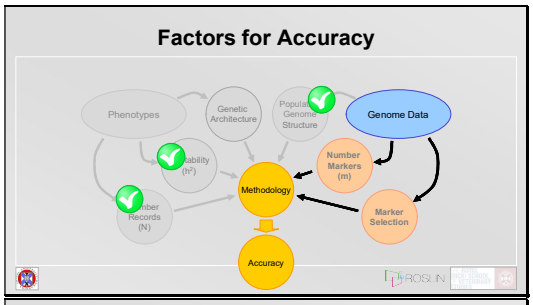
Me is an 'effective number' of loci – the genome behaves <u>as if</u> it were Me unlinked loci. Historically the reliance of quantitative genetics on the infinitesimal model had led it to being criticised as 'bean bag genetics' (Kempthorne, I think), as a black box, unconcerned with what was inside it. There is a real sense that Me is the number of beans in the bean bag.

There are predictive formulae which are functions of Ne and the number of chromosomes, but these are beyond the course. However they indicate that while pedigree Labrador dogs may have ~650, humans will have ~100,000.

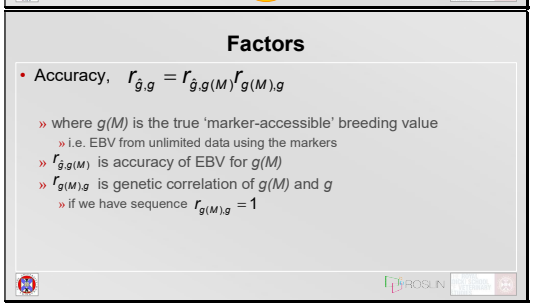
So continuing the idea of 'information', the information per bean, $\lambda = h^2N / Me$.

Then a very good approximation is $r_{\hat{g},g} = \sqrt{\lambda / (1+\lambda)}$.

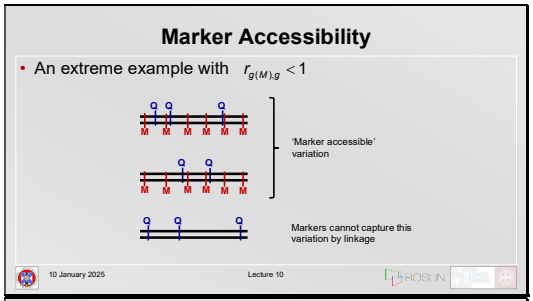




A major feature of genomic data is the number of markers and their distribution and the selection of markers. It is possible to give some important perspectives.

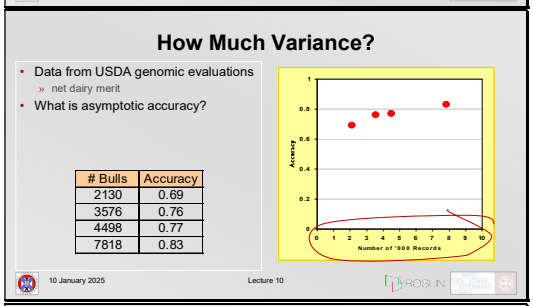


Firstly, most obviously, the markers have to be dense and ubiquitous, so that all loci have markers in their neighbourhood. If this is not the case then some of the variation may not be accessible to the markers. Accuracy can then be viewed as the product of the accuracy of predicting the marker-accessible genome ($r_{\hat{g},g(M)}$) and the genetic correlation between the marker accessible genome and full genome ($r_{g(M),g}$). Strictly these will be trait dependent, but one might imagine these for a typical complex trait like BMI or height.

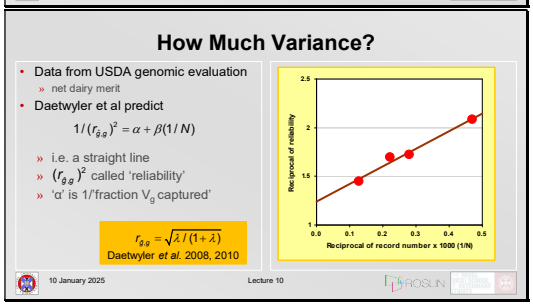


The value of $r_{g(M),g}$ would be the highest accuracy for any set of markers and a trait, and it would be hoped that it is always close to 1. If one has sequence data one might be able to assume that $r_{g(M),g} = 1$.

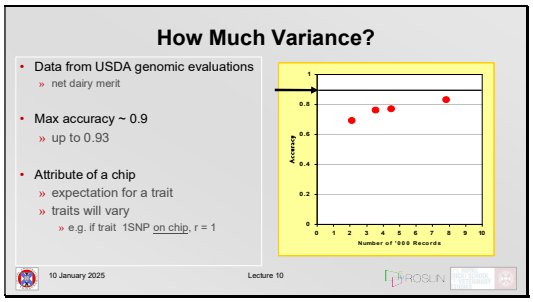
An obvious case where $r_{g(M),g} < 1$ is shown, where the the markers do not cover all the chromosomes.



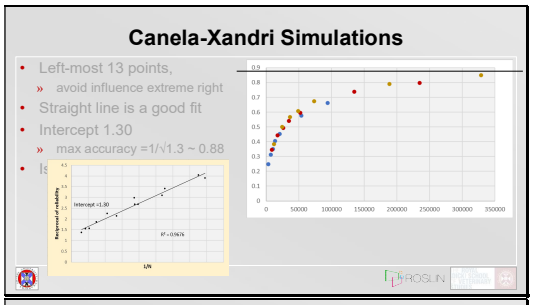
Here is a small example with progeny tested bulls. Many of these bulls get very accurate estimate of their breeding values obtained empirically from the large number of offspring they have. Using these, the value of $r_{\hat{g},g(M)}$ can be estimated for a standard 50k SNP chip used in the industry by looking at how the accuracy changed with increasing amounts of data.



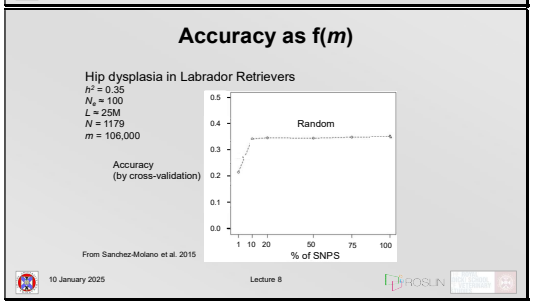
The formula from Daetwyler et al. (2008) can be modified to show that the reciprocal of the squared accuracy is predicted to have a straight-line relationship with 1/N where N is the number of records. As N becomes very large and the accuracy will tend to $r_{g(M),g}$, 1/N will tends to 0, and the intercept to an estimate of $1/r^2_{g(M),g}$.



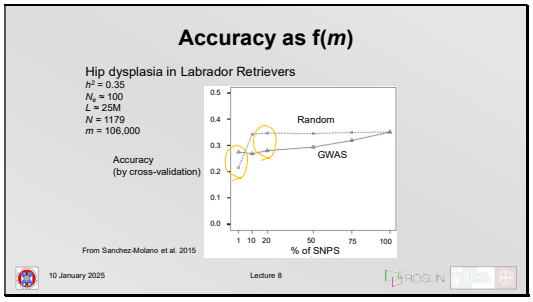
In this data the intercept was 1.244 (s.e. 0.053), and not 1 which would be the case if all the variance was being captured by this chip. The observed intercept suggests $r_{g(M),g} \approx 0.9$ for a typical complex trait.



The same can be done with the simulations of the human genome by Caneli-Xandri et al. (2015). With simulated data the true additive genetic merits are known so the accuracy is well-estimated. The full data shows some curvilinearity, but with the 13 points with the largest N, hence small 1/N, the straight-line fit is particularly good, and so will give the best estimate of the intercept. The intercept of 1.30 estimates the maximum accuracy as $r_{g(M),g} \approx 0.88$.

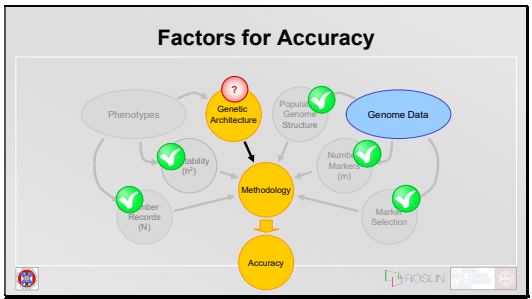


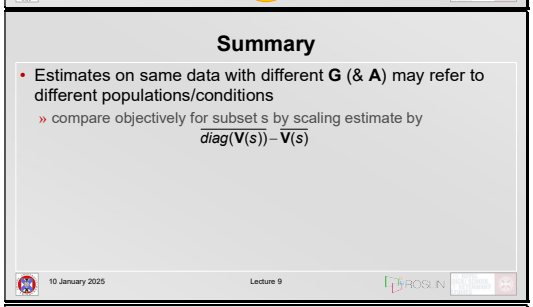
How many markers are needed for GBLUP? It is quite a common finding across species that the accuracy of GBLUP as a function of the number of markers included in the calculation of a Van Raden **G** very quickly reaches a plateau. (A caution: much of the evidence comes from SNP chip loci, which are not a random sample of SNP in the genome, biased towards intermediate frequency and do not include the causal variants). In the example of the Labrador dogs, cross-validation showed the number of loci needed to reach the plateau was 20k to 25k randomly chosen SNP chip loci.



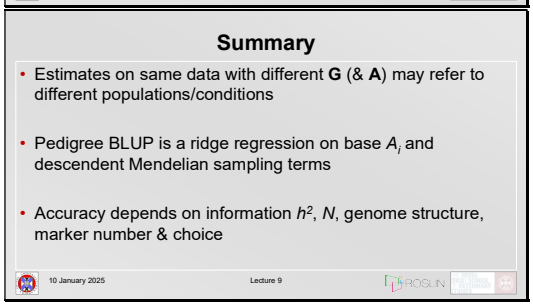
It is worth noting that a random choice of SNP was important (Sanchez-Molano et al. 2015). A comparison was made with selecting SNP after ranking by GWAS hits. With very low (sub-optimal) SNP density the GWAS hits do comparatively well, but then very quickly become worse than random, only reaching parity when all SNP are selected. This is because the very top of the GWAS ranking is enriched with SNP that are in true LD with causal variants – which are predictive in a validation set, whereas lower down the ranking SNP effects are overestimated and are possibly predictive of hidden structure in the data. Consequently, the estimates add noise rather than precision in predicting the breeding values in the validation set, i.e. not in the training set. Other criteria such as spacing, MAF may be important but beware!

So having covered the other items, what is left is the 'Genomic Architecture'. Until now it has generally been assumed that the effects of the causal variants are Normally distributed, or approximately so. This is a property of the trait, not a property of the species genome which is determined by LD and has been loosely modelled by Me. How much does this assumption of normal QTL effects matter, and first of all how can we get tractable models to address this? This will be the topic for much of Lecture 10.





The variance component estimated does not necessarily apply to populations of interest, and different models may be estimating variances for different, sometimes virtual, populations. However objective comparisons can be made by considering the variance within a desired target population by using $\overline{diag(\mathbf{V}(s))} - \overline{\mathbf{V}(s)}$, the mean of the diagonal elements minus the average of all the matrix.



All models we have used, including **A** have an underlying ridge regression model,

Finally the accuracy is a function of the h^2N but will be affected by the genomic structure of the species whereby species with high Me, with LD extending only over short distances, will require more records to achieve the same accuracy for a trait. Accuracy will increase with marker numbers up towards a plateau. Selection of SNPs other than random has pitfalls, especially if selection is made on the basis of GWAS studies.